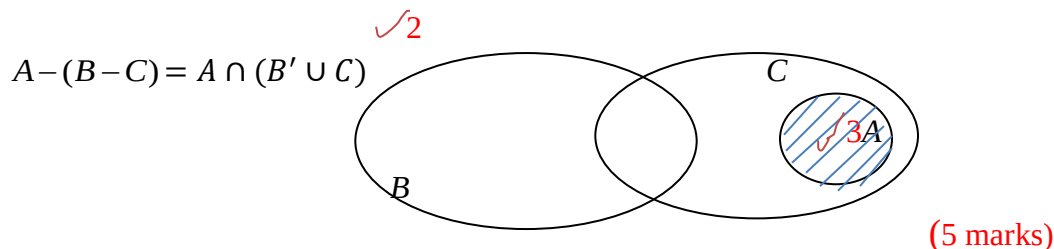


The University of Zambia
School of Natural Sciences
Department of Mathematics & Statistics
MAT1100 Foundation Mathematics Quiz 1
Solutions

1. (a) In the Venn diagram below shade the set $A - (B - C)$:



- (b) A binary operation $*$ is defined on $\mathbb{R}$ by $p * q = (p - q)^2 + 2pq$.

- (i) Determine whether the operation is commutative.

$$p * q = (p - q)^2 + 2pq$$

$$q * p = (q - p)^2 + 2qp = [-(p - q)]^2 + 2pq \quad \checkmark 2$$

$$= (p - q)^2 + 2pq = p * q$$

Therefore, the operation is commutative on $\mathbb{R}$. ✓1 (3 marks)

- (ii) Evaluate $2 * (-3 * 1)$.

$$(-3 * 1) = (-3 - 1)^2 + 2(-3)(1) = (-4)^2 - 6 = 16 - 6 = 10 \quad \checkmark 1$$

$$\therefore 2 * 10 = (2 - 10)^2 + 2(2)(10) = (-8)^2 + 40 = 64 + 40 = 104. \quad \checkmark 1$$

(2 marks)

2. (a) Express, in simplest form, the number $2.13\overline{42}$ in the form $\frac{p}{q}$, where

$$p, q \in \mathbb{Z}.$$

$$\text{Let } p = 2.13\overline{42}. \text{ then } 100p = 213.\overline{42} \text{ and } 10000p = 21342.\overline{42}$$

$$\Rightarrow 10000p - 100p = 21342.\overline{42} - 213.\overline{42}$$

$$\Rightarrow 9900p = 21129$$

$$\therefore p = \frac{21129}{9900} = \frac{7043}{3300}$$

(5 marks)

- (b) Given that $z = \frac{(5-2i)^2}{i-2}$, express z in the form $a + ib$, where $a, b \in \mathbb{R}$.

$$z = \frac{(5-2i)^2}{i-2} = \frac{25-20i-4}{i-2} = \frac{21-20i}{-2+i} \times \frac{-2-i}{-2-i} = \frac{-42+40i-21i-20}{5}$$

$$= \frac{-62+19i}{5} = -\frac{62}{5} + \frac{19}{5}i \quad \checkmark 2$$

(5 marks)

3. (a) Complete the square of the function $f(x) = 6 - x - x^2$. Hence, state its maximum or minimum value.

$$f(x) = 6 - x - x^2 = -(x^2 + x - 6) = -\left(x^2 + x + \frac{1}{4} - \frac{1}{4} - 6\right)$$

$$= -\left[\left(x + \frac{1}{2}\right)^2 - \frac{25}{4}\right] = -\left(x + \frac{1}{2}\right)^2 + \frac{25}{4}. \quad \checkmark 3$$

Hence, the maximum value of the function is $f\left(-\frac{1}{2}\right) = \frac{25}{4}. \quad \checkmark 2$

(5 marks)

- (b) The equation $2x^2 - x + 3 = 0$ has roots α and β . Find the value of $\alpha^2 + \beta^2$.

$$a = 2, b = -1, c = 3 \quad \checkmark 1$$

$$\alpha + \beta = \frac{-b}{a} = \frac{-(-1)}{2} = \frac{1}{2}, \alpha\beta = \frac{c}{a} = \frac{3}{2}. \quad \checkmark 1$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(\frac{1}{2}\right)^2 - 2\left(\frac{3}{2}\right) = \frac{1}{4} - 3 = \frac{1-12}{4} = -\frac{11}{4}. \quad \checkmark 1$$

(5 marks)

4. (a) (i) Determine whether the function $f(x) = 2x - 3x^3$ is even or odd or neither even nor odd.

$$f(-x) = 2(-x) - 3(-x)^3 = -2x + 3x^3 = -(2x - 3x^3) = -f(x) \quad \checkmark 1$$

$$\Rightarrow \text{the function is odd.} \quad \checkmark 1$$

(3 marks)

- (ii) Determine whether the function $f(x) = \frac{2x-1}{x+1}$ is one-to-one or not.

$$f(x_1) = f(x_2) \Rightarrow \frac{2x_1-1}{x_1+1} = \frac{2x_2-1}{x_2+1} \quad \checkmark 1$$

$$\Rightarrow (2x_1 - 1)(x_2 + 1) = (x_1 + 1)(2x_2 - 1)$$

$$\Rightarrow 2x_1x_2 + 2x_1 - x_2 - 1 = 2x_1x_2 - x_1 + 2x_2 - 1$$

$$\Rightarrow 3x_1 = 3x_2 \Rightarrow x_1 = x_2. \quad \checkmark 1$$

$$\therefore \text{the function is one-to-one.} \quad \checkmark 1$$

(3 marks)

- (b) Factorize the polynomial $f(x) = x^3 + x^2 - 4x - 4$, and hence sketch the graph of the function $y = f(x)$ indicating the x - and y - intercept.

$$f(-1) = (-1)^3 + (-1)^2 - 4(-1) - 4 = -1 + 1 + 4 - 4 = 0$$

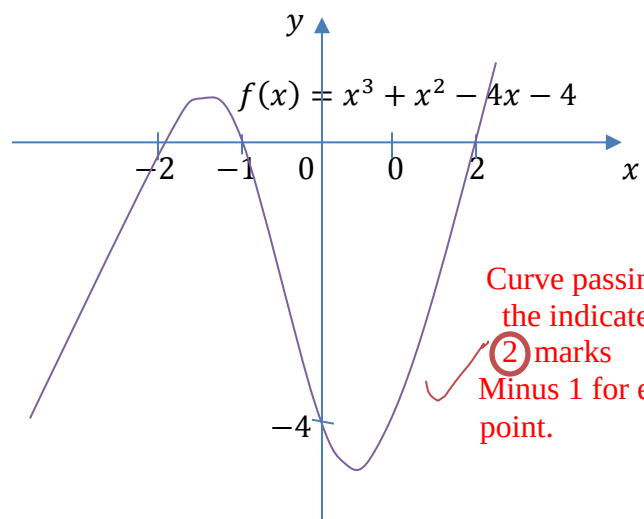
$$\Rightarrow x + 1 \text{ is a factor of } f(x). \quad \checkmark 1$$

$$\begin{array}{r} -1 \overline{) \quad 1 \quad 1 \quad -4 \quad -4} \\ \underline{ -1 \quad 0 \quad 4} \\ 1 \quad 0 \quad -4 \quad 0 \end{array}$$

$$\Rightarrow f(x) = (x + 1)(x^2 - 4) = (x + 1)(x + 2)(x - 2) \quad \checkmark 1$$

x – intercepts are $-1, -2, 2$,

y – intercept is -4 .



Curve passing through
the indicated points
② marks
Minus 1 for each wrong
point.

(4 marks)